

LiPON — phosphate-oxynitride applicability limit

A chemically constrained network provides the most stringent pretrained-potential test

1 Literature

Lacivita et al. (2018)
Neutron / IR + AIMD
Seth et al. (2025): NequIP

2 Structure

$\text{Li}_{47}\text{P}_{16}\text{O}_{56}\text{N}_5$
124 atoms; Preparation B
Apical + bridging N

3 MD protocol

600 / 900 / 1200 / 1500 K
50 ps equilibration
300-600 ps production

4 Analysis

MSD, $D(T)$, σ_{NE} and E_a
P-O / P-N / Li RDF
Network-topology checks

5 Conclusion

Thermal trend reproduced
D strongly overestimated
Transferability limit exposed